

Digital Human Live Commerce Case Study

Unmanned, Script-Driven AI Livestreaming at Baidu

Note: This case is written **qualitatively** — the digital-human project's internal result metrics (PV/CVR/GMV) were never publicly disclosed, so it is framed around the 0-to-1 capability, the reliability-first judgement and the Du Xiaoxiao IP landing rather than performance numbers.

One-line judgement: For live commerce, the bottleneck was never content — it was the cost and availability of a human anchor. The winning product was not a more lifelike avatar, but an unmanned, script-driven live room that a non-technical merchant could set up and trust to run 24/7.



Role & Scope

Role: Lead Product Manager (owned the product end-to-end)

Product: Du Xiaoxiao Digital Human Live Commerce — an AI digital-human livestreaming platform that lets merchants run automated, always-on live selling using Baidu's Du Xiaoxiao AI persona.

Scope: Problem framing, product strategy, B-end (merchant setup) and C-end (viewer experience) design, real-time interaction model, applied-AI scoping, cross-functional delivery with engineering, algorithm, design and operations.

Context: Livestream commerce boom + pandemic-era pressure on offline retail; Baidu building out its live-commerce platform (a £2M+ daily GMV business) and looking to give its Du Xiaoxiao AI persona a real commercial landing.

Context

Livestream shopping had become one of the fastest-growing commerce formats, but it ran on a fragile, expensive input: **human anchors**. A live room needs a person who is on-camera, on-script and on-brand — which means real anchors are costly, hard to schedule, impossible to run around the clock, and out of reach for small and micro merchants. During the pandemic this gap widened: many merchants, and even farmers selling produce, needed to sell online but had no anchor, no studio and no budget.

At the same time, Baidu had a well-known AI persona, Du Xiaoxiao, that needed commercial, revenue-generating use cases rather than only marketing moments.

The initial framing was tempting but wrong: *"build a more realistic AI streamer."* I reframed it around the real constraint.

Problem

The core problem was not *"livestream rooms need better-looking hosts."* It was:

"How does a merchant with no anchor, no studio and no budget run a credible live room that sells — reliably, and around the clock?"

That reframes the product from an **avatar-quality problem** into an **operating-cost and reliability problem**. The AI digital human is only valuable if it removes the human bottleneck *"without"* making the live room feel broken, off-brand or untrustworthy to shoppers.

Insight

Merchants don't want a magical AI. They want a live room that **runs without them, stays on-brand, and doesn't embarrass them in front of customers**.

That shifted the design goal from "impressive generation" to **"reliable, controllable, merchant-operable automation"** — an AI product judged by uptime, brand-safety and conversion, not by how lifelike a single clip looks.

Strategic Decisions

1. **Script-driven, not free-form.** Rather than let a generative model improvise the whole broadcast, I structured the live room around configurable **script modules** — opening/closing, marketing, product explanation, fixed interactions, and real-time interactions. A merchant edits and publishes these before going live, and the system renders them into a continuous audio-video stream. This traded spontaneity for **reliability, brand control and merchant setup without any technical skill** — the qualities a commerce product actually needs.

2. **A real-time interaction engine on top of the script.** To avoid a "dead" pre-rendered feel, the live room reacts to viewer behaviour — entry, likes, follows, gifts, and orders each trigger the matching bound action clip plus text-to-speech, composed live into the stream (e.g. "Welcome, [user] — thanks for the like!"). This kept the room feeling live and responsive while staying fully automated.

3. **Ship the answerable version of Q&A; first.** True real-time semantic Q&A; (voice + matching body language) was the ideal, but the highest-risk piece. I scoped Phase 1 to fall back to a **text-bubble reply** when full voice-action response wasn't ready — so the product shipped and sold, instead of blocking on the hardest AI problem.

4. **Leverage an existing, trusted IP (Du Xiaoxiao).** Rather than a generic avatar, binding the room to a persona users already recognised gave it warmth, personality and stickiness — and gave Baidu's AI persona a genuine commercial home.

Trade-offs

The central trade-off was **control vs. spontaneity**.

A script-driven digital human is less "wow" than a fully generative one — it won't freestyle. But in a commerce setting, freestyling is a liability: it risks off-brand claims, wrong product details and moments no merchant would sign off on. Choosing a controllable, configurable system meant the product was **trustworthy enough to actually sell under a brand's name**, and simple enough that a non-technical merchant could operate it. The bet was that reliability and brand-safety would matter more than novelty for real GMV — which is the judgement I'd defend.

The second trade-off was **automation vs. human warmth**. Fully unmanned live risks feeling robotic; the real-time interaction engine and a familiar persona were the deliberate counter-weights.

Outcomes & Impact

- Built Baidu's digital-human live-commerce capability **0-to-1** — a script-driven, fully unmanned live room covering the whole selling loop: opening, product explanation, lucky-draws, coupons, real-time interaction and Q&A.;
- Enabled **24/7 unmanned live selling**, removing the anchor, studio and manpower cost that previously gated a live room — extending live commerce to merchants who couldn't staff an anchor, including small/micro and agricultural sellers.
- Landed the **Du Xiaoxiao brand-IP digital human** as a commercial, revenue-generating selling scenario, not just a marketing moment — connecting it with quality supply and digital-collectible tie-ins.
- Defined the **"24H stand-in anchor"** and **"human + digital-human co-hosting"** models from competitive research (benchmarking Miaobo, Silicon-based Xiaoboxiu, and Tencent Zhiying / Huiboxing), giving the platform a differentiated path.

What I Learned

- **In commerce AI, controllability beats spontaneity.** The product succeeded because merchants could trust and operate it — not because the avatar was the most lifelike. Judging AI features by uptime, brand-safety and conversion (not by demo wow-factor) is the discipline that shipped it.
- **Apply AI where it removes a real cost, not for novelty.** The value wasn't "an AI streamer" — it was deleting the human bottleneck that kept live commerce out of most merchants' reach.
- **Meet non-technical users where they are.** Turning a hard AI capability into a *configurable script* a merchant could set up themselves is what made the automation actually get adopted.
- **Scope around the riskiest assumption.** Falling back to text-bubble Q&A; in Phase 1 let the product ship and learn, instead of stalling on the hardest interaction problem.